

CR-002: CSV-Based Student Data Import and Add Student Form Auto-Fill

1. Change Request Details

Field	Details
Change Request ID	CR022
Title	CSV-Based Student Data Import and Add Student Form Auto-Fill
Priority	High
Difficulty	Medium
Change Type	Frontend and backend Enhancement
Affected Version	Student Management System v1.1.0
Dependency	Existing Add New Student form and student creation workflow

2. Business Requirement

Administrators currently need to enter student information manually into the Add New Student form. When multiple student records are available in a CSV file, manually entering each field can be time-consuming and may increase the possibility of data-entry errors.

The system should provide a CSV upload facility within the Add New Student form so that student information contained in a valid CSV file can be read and used to automatically populate the corresponding form fields. The administrator should be able to review and edit the populated information before submitting the student record.

3. Current Behaviour

- Administrators enter student information manually through the existing Add New Student form.
- There is no CSV file upload facility for populating the student form.
- Student details must be entered field by field.
- The existing student creation workflow is used to submit the completed form.

4. Requested Change

- Add a CSV file upload control to the Add New Student form.
- Allow the administrator to select and upload a CSV file containing student information.
- Read the CSV file on the client side and map CSV column headers to the corresponding Add New Student form fields.
- Automatically populate the form fields using the CSV data.
- Support the student information fields that already exist in the Add New Student form.
- Validate the uploaded file type and CSV structure before populating the form.
- Display a clear validation message when the CSV file is invalid, empty, malformed, or missing required columns.
- Allow the administrator to review and manually edit the populated values before submission.
- Do not automatically create a student record immediately after CSV upload.
- Use the existing Add New Student submission workflow after the administrator confirms the data.
- Preserve the existing manual data-entry functionality.
- Preserve existing authentication, authorization, validation, and student creation behaviour.

5. Scope

In Scope

- CSV upload control in the Add New Student form.
- CSV file reading and parsing.
- CSV column-to-form-field mapping.
- Automatic population of existing form fields.
- CSV format and required-column validation.
- Invalid CSV and validation error messages.
- Review and editing of imported values before submission.
- Use of the existing student creation workflow.
- Preservation of manual student data entry.

Out of Scope

- Changes to the student database structure.
- Bulk direct database insertion from CSV.
- Automatic student creation immediately after file upload.
- New student-management API endpoints unless technically required by the existing architecture.
- Changes to administrator authentication or authorization.
- Changes to unrelated student-management features.

6. Expected / Potentially Affected Files

- Existing Add New Student page/component.
- Existing dashboard stylesheet or form stylesheet.
- A new CSV parsing/helper component or utility if required.
- Existing student creation API integration only if required by the implementation.

Students may modify additional files if required. Any additional files must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

- FR-001: A CSV upload control shall be available in the Add New Student form.
- FR-002: The system shall accept a valid CSV file for importing student form data.
- FR-003: The system shall identify CSV columns using the defined column headers.
- FR-004: CSV values shall be mapped to the corresponding existing Add New Student form fields.
- FR-005: Successfully parsed values shall automatically populate the relevant form fields.
- FR-006: The administrator shall be able to edit imported values before submission.
- FR-007: The system shall display a clear error when the uploaded CSV is invalid or cannot be parsed.
- FR-008: The system shall identify missing required CSV columns or required student information.
- FR-009: CSV upload shall not automatically submit or create the student record.
- FR-010: The existing Add New Student submission workflow shall remain functional.
- FR-011: Manual entry shall continue to be available when no CSV file is used.
- SEC-001: Existing administrator authentication and authorization shall remain unchanged.
- SEC-002: Uploaded CSV content shall be validated before being used to populate the form.
- SEC-003: The implementation shall not use unsafe HTML rendering for imported CSV values.
- SEC-004: Imported data shall not bypass the existing server-side validation and authorization controls.

8. Acceptance Criteria

- AC-001: Administrators can select and upload a valid CSV file from the Add New Student form.
- AC-002: Valid CSV columns are correctly mapped to the corresponding form fields.
- AC-003: Imported values are automatically displayed in the Add New Student form.
- AC-004: The administrator can edit imported values before submitting the form.
- AC-005: Invalid, empty, or malformed CSV files display a clear error message.
- AC-006: Missing required CSV columns or required data are clearly identified.
- AC-007: Uploading a CSV does not automatically create a student record.
- AC-008: Submitting the completed form continues to use the existing student creation workflow.
- AC-009: Existing manual student entry continues to work.
- AC-010: Existing administrator authentication, authorization, validation, and CRUD functionality continue to work.

9. Expected Outcome

Administrators can upload a properly structured CSV file and use its student information to automatically populate the Add New Student form. This reduces repetitive manual data entry while keeping the administrator in control of reviewing, editing, and submitting the student information. Existing student creation, authentication, authorization, and validation functionality remain unchanged.

10. Implementation Report Requirement

- Implementation summary.
- CSV format and column mapping used.
- Affected files.
- Important technical and security changes.
- CSV validation and error-handling details.
- Test cases and results.
- Acceptance criteria verification.
- Relevant screenshots/evidence.
- Final status and outstanding issues.